

Graph inputs

Helper results

Blueprint hypotheses

Conclusions

Scalar graph block maps
Prop. F.1
order + constant shifts

Prop. G.1
order + shifts $\Rightarrow$
variation non-expansiveness

Ψ non-expansive
in the first block quotient

Positive cost +
reverse edges
Prop. F.2
 H_γ explicit

$\|\log(x^*/z)\|_\infty \leq H_\gamma$
 κ_1 finite, def. (5)

Dual helper
Prop. 4.1
 $U_\gamma = 2\kappa_1(\|C\|_\infty + \gamma H_\gamma)$

First-block or-
bit and optimum
quotient radius $\leq U_\gamma$

Paths: $\kappa_1 \leq D_{\text{hop}}$
Prop. F.3
 $b_2 = 0, \|A\|_{1 \rightarrow 1} = 2$

Pairing bound $\langle b, u \rangle \leq P_\gamma$
not independent gauges
graph: $P_\gamma = \|q\|_1 U_\gamma$

Primal helper
Prop. 4.2
 $X_\gamma = P_\gamma/\gamma + \sum_i z_i e^{-C_i/\gamma}$

Full and half-step
masses $\leq X_\gamma$

Normalized graph lift
cost $(W/2, W/2)$
parameter $\theta = \gamma/2$

Lemma A.3
bounded-mass Pinsker
 $\eta_\gamma = 1/(2X_\gamma)$

Thm. 3.1
 $\Delta_k \leq 8X_\gamma U_\gamma^2 a^2/(\gamma k)$
 $a = \|A\|_{1 \rightarrow 1}$

Graph value estimate
Thm. 5.1
fixed data: $O(p\epsilon^{-3})$

Bias envelope B_0
Lemma B.1
 $0 \leq F_\gamma^* - F_0^* \leq \gamma B_0$

Thm. 3.2
 $\gamma = \epsilon/(2B_0)$
 $K \geq 32X_\gamma U_\gamma^2 a^2 B_0/\epsilon^2$

